ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
GUWAHATI

Course Structure and Syllabus

(From Academic Session 2018-19 onwards)

B.TECH
INDUSTRIAL AND PRODUCTION ENGINEERING
8th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure

(From Academic Session 2018-19 onwards)

B. Tech 8th Semester: Industrial and Production Engineering

Semester VIII/ B. TECH/IPE

Sl. No.	Sub-Code	Subject	Hours per Week			Credits	Marks	
			L	T	P	C	CE	ESE
Theory								
1	IPE181801	Non-Traditional Production Process	3	0	0	3	30	70
2	IPE1818PE2*	Program Elective –2	3	0	0	3	30	70
3	IPE1818PE3*	Program Elective -3	3	0	0	3	30	70
4	IPE1818OE2*	Open Elective- 2	3	0	0	3	30	70
Practical								
1	IPE181822	Project -2	0	0	12	6	100	50
2	IPE181823	Grand Viva Voce-II	0	0	0	1	-	50
TOTAL			**12**	**0**	**12**	**19**	**220**	**380**
Total Contact Hours per week : 24								
Total Credits: 19								

Program Elective-2 Subjects		
Sl. No.	**Subject Code**	**Subject**
1	IPE1818PE21	Industrial Marketing
2	IPE1818PE22	System dynamics and Modelling
3	IPE1818PE2*	Any other subject offered from time to time with the approval of the University

Program Elective-3 Subjects		
Sl. No.	**Subject Code**	**Subject**
1	IPE1818PE31	Industrial Statistics
2	IPE1818PE32	Computer Aided Design and Manufacturing
3	IPE1818PE3*	Any other subject offered from time to time with the approval of the University

Open Elective-2 Subjects		
Sl. No.	**Subject Code**	**Subject**
1	IPE1818OE21	Materials Management
2	IPE1818OE22	Optimization in Operations Research
3	IPE1818OE23	Management Information Systems
4	IPE1818OE2*	Any other subject offered from time to time with the approval of the University

Detailed Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE181801	**Non-Traditional Production Process**	**3-0-0**	**3**

Course Outcomes (COs): At the completion of the course the student will be able:
1. To apply the non-traditional production processes in industrial applications.
2. To implement the various abrasive machining concept and processes in practice.
3. To identify and apply the right electrochemical processes, EBM, PAM, LBM, micro-finishing and HERF processes in real life operations.
4. To select most suitable process for a products and analyze economic analysis of the non-traditional machining processes.

MODULE 1:
Introduction to the new methods of Production, Need and capability analysis of the various processes; classification and selection.

MODULE 2:
Abrasive processes of machining: Abrasive Jet Machining (AJM), Water Jet Machining (WJM) and ultrasonic Machining (USM). Equipments and the processes; advantages and Application.

MODULE 3:
Chemical Machining (CHM) and Electro-chemical, Machining (ECM): Principle and steps in the process; merits, demerits and application. Electro-chemical deburring and Honing. Electro-Chemical Grinding (ECG). Electrical Discharge Machining (EDM). Principle of the process; the EDM machine and tools. Field of application. Comparison with ECM.

MODULE 4:
The principle and process of Electron Beam Machining (EBM), Plasma Arc Machining (PAM) and Laser Beam Machining (LBM). Comparative study of the processes and applications.

MODULE 5:
High Energy Roll Forming (HERF) or High Velocity Forming (HVF) -- Explosive Forming, Electrohydraulic Forming, Magnetic Forming and Pneumatic Mechanical Forming. Principle, steps and application.

MODULE 6:
Selection of the most suitable process for a product. Economic analysis of the non-traditional machining processes.

MODULE 7:
Micro-finishing Processes Different processes – Grinding, Lapping, Honing and super-finishing – Polishing, Buffing, Tumbling and Burnishing.

Textbooks/ Reference Books:

1. Elements of Elements of Workshop Technology (Volume – 1 & 2) - Author: S. K. Hajra Choudhury, Nirjhar Roy, A. K. Hajra Choudhury – Media Promoters
2. Manufacturing Science Manufacturing science- Amitabha Ghosh, Asok Kumar Mallik – Prentice Hall India
3. Manufacturing Technology: Metal Cutting and machine Tools, Vol – 2 – P.N. Rao, Mc

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1818PE21	Industrial Marketing	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:

1. To apply the marketing concept in real life situations.
2. To implement the various market research processes in practical field.
3. To identify and apply the right advertising and sales promotion techniques to establish a business.
4. To apply the right pricing strategy to run a business effectively.

MODULE 1: Basic Consideration

Concept, nature and scope of Industrial marketing management and marketing functions, Planning: Annual Plan; element of marketing mix, Factors affecting marketing mix – variable relating to market.

MODULE 2: Market Segmentation

Definition and Importance – types – product oriented market segmentation, criteria for segmenting a market. Marketing Information system, Information needs of Marketing Executive – Information based decision making.

MODULE 3: Marketing Research

meaning, aims and objectives – organization of research groups – sources of marketing research and technique: Questionnaire and marketing research, steps designing and questionnaire. Marketing research and market research.

MODULE 4: Consumer behaviour

demand for goods, buying decision process, buyers behaviour Models. Buying motives, Motivation research Techniques: Buyer and Seller relationship – decision making theories in brief – decision making model and procedure. Marketing Environment.

MODULE 5: Advertising and Sales promotion

Role and importance of advertising in selling – social responsibilities in advertising – advertising methods – product branding, packaging and labelling. Sales promotion management – Personal selling – sales forecasting. Industrial marketing control.

MODULE 6: Pricing: Strategy and decision

Conditions affecting Price – Pricing policies: Cost analysis and pricing – pricing methods -- price discrimination. Evaluation of Marketing performance.

Textbooks/ Reference Books:

1. Marketing Research (King's Book) ------ M.M. Verma and R.K. Agarwal.
2. Element of Marketing Management (KedarNath, RamNath) ---- P. Kumar.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1818PE22	**System Dynamics and Modelling**	**3-0-0**	**3**

Course Outcomes (COs): At the completion of the course the student will be able:
1. To develop the concept of system dynamic paradigm and correlate them in real life systems.
2. To implement the various System Dynamics modelling processes in practical life.
3. To identify and apply the System order and feedback systems in system dynamics causal loop.
4. To construct the system dynamics models to depict the real life systems and analyze the system to develop suitable policies.

MODULE 1: Introduction
The system dynamics paradigm, system dynamics viewpoints, problem solving using system dynamics

MODULE 2: Introduction of system
System Concept, system theories, system models, System thinking

MODULE 3: System Dynamics modelling
physical flows, level and rate variables, information flow, flow diagrams, delays, smoothing of information, table function and multipliers. Causal Loop diagramming, diagramming approach, causal links and its justification, examples of. Causal Loop diagramming

MODULE 4: System order and feedback systems
order of a system, linearity of system, feedback systems- positive and negative feedback system with first order and second order, complex feedback systems, steps of system dynamics modelling.

MODULE 5: Policy planning through system dynamics
Building and simulating system dynamics models, Model validation, Policy design, Algorithms for resource allocation and dynamic policy option selection

MODULE 6: Tools of system dynamics modelling
diagrams, equations, and software. Software Packages: STELLA, VENSIM. Practice sessions in computer with standard examples.

Textbooks/ Reference Books:

1.Introduction to System Dynamics Modelling, Pratap K J Mohapatra, Purnendu Mandal, Madhab C Bora, Orient Longman Limited, New Delhi; Universities Press (india) Ltd., Hyderabad;
2.John Sterman, Business Dynamics: Systems Thinking and Modeling for a Complex World, Irwin/McGraw-Hill (2000)
3.J. W. Forrester, Industrial Dynamics, Cambridge MA: Productivity Press (1961)
4.Sushil, System Dynamics: A Practical Approach for Managerial Problems, Wiley Eastern (1993)

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1818PE31	Industrial Statistics	3-0-0	3

Course Outcomes (COs): At the completion of the course the student will be able:
1. To apply the concept of Statistical tests of significance, correlation and regression analysis and Auto-correlation in real life.
2. To implement the various Statistical Quality Control and Acceptance Sampling plans to make tighter plans for quality inspection.
3. To identify and apply the Analysis of variance & Design of Experiments in practical problems.
4. To solve complex stochastic problems in engineering.

MODULE 1:
Basic Statistical Concept Method of collection and presentation of data – variables and attribute. The frequency distribution and its graphical representation: normal distribution. Measurement of Central tendency and dispersion.

MODULE 2:
Probability Definition and basic laws of probabilities, conditional probability, random variables, binomial, Poisson's & normal distribution (only properties and applications) sampling distributions: statistical hypotheses. Statistical tests of significance, correlation and regression analysis. Auto-correlation.

MODULE 3:
Statistical Quality Control charts for variables -- X & R chart and control charts for attributes -- P and C charts.

MODULE 4:
Acceptance Sampling Introduction – Methods, The O.C. Curve: Quality indices for acceptance sampling plans -- design of an acceptance plan.

MODULE 5:
Analysis of variance & Design of Experiments One way & two-way classification, Basic Experimental design: Randomised block, Latin square, Factorial 22 experiment.

MODULE 6:
Introduction to stochastic problems in engineering.

Textbooks/ Reference Books:

1. Statistical Methods ---- an introduction by J. Medhi
2. Fundamentals of Statistics Vol – II by A.M. Goon, M.K. Gupta and B.D. Gupta (Chapters 20, 27)
3. Fundamentals of Applied Statistics by S.C. Gupta and V.K. Kapoor (Chapter 5 & 6)

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1818PE32	**Computer Aided Design and Manufacturing**	**3-0-0**	**3**

Course Outcomes (COs): At the completion of the course the student will be able to:

1. Design 2D and 3D models using CAD software for different practical applications.
2. Apply the concept of geometric transformation and manipulate the design of curves.
3. Apply the concept of Numerical Control and Part Programming on NC and CNC Systems.
4. Justify the requirements and changes adapted in FMS and implement the group technology.
5. Develop plan in Management and implementation of Computer Integrated Manufacturing Systems.

MODULE 1: Introduction to CAD / CAM

Definition of CAD/CAM and its importance, Design process, Computer system and related Technology, Application of computers for design, CAD/CAM hardware and System, Computer Languages. CAD/CAM software, Definition of system software and application software.

MODULE 2: Geometric Transformations, Curves and Surfaces

2 D Geometric Transformations-Translation, Scaling, Shearing, Rotation & Reflection Matrix representation, Composite transformation, 3 D transformations. Interpolation and approximation of curves, Concept of parametric and non-parametric representation of curves, Curve fitting techniques, definitions of cubic spline, Bezier, and B-spline, Parametric surfaces.

MODULE 3: Computer Graphics and Modelling

Software Configuration, Geometric and solid Modelling, Building block for Solid, Wire frame Modelling – Finite element Analysis. Database Management – Databank Concept, CAD/CAM Databases – Databank Information Storage and retrieval, Data life Cycle – integrated data Processing.

MODULE 4: Numerical Control and Part Programming

Growth development and Component in NC system – Operation of a NC machine tools system – Co-ordinate system – Binary System – Basic motions of NC System. CNC system – CNC-DNC and adaptive Control. Justification and Economics – Part programming and Computer Languages.

MODULE 5: Group Technology and FMS

Concept of Group Technology, Process Planning and Group technology, benefits of Group Technology. Definition and concept of of FMS -- Work Stations – Planning, different Types and Technology required for FMS – justification.

MODULE 6: Computer Integrated Manufacturing Systems (CIM)

Elements of CIM – approach to CIM and Steps – Planning and Implementation – Material requirement planning, Capacity Planning – Shop floor Control. Role of Management in CIM

<u>REFERENCE BOOKS:</u>

1. CAD/CAM (Dhan Pat Rai & Sons.) ------ S. Kumar and A.K. Jha
2. Computer Integated Manufacturing (PHI) ------ S. K. Vajpayee.
3. Mechatronics, HMT Ltd., (Tata McGraw Hills) .

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1818OE21	**Materials Management**	**3-0-0**	**3**

Course Outcomes (COs): At the completion of the course the student will be able:
1. To apply the Integrated Material management in ordering or purchasing materials.
2. To implement the Material Requirement Planning (MRP) to design MRP system.
3. To identify and apply the Inventory Control concept to design optimum order size.
4. To apply concept of store keeping and purchasing in practical life.

MODULE 1: Definition, Importance and Objective
Integrated Material management; Organization and control; Codification – requirement, Types and method. Simplification and standardization.

MODULE 2: Material requirement Planning (MRP)
Principle, Pre-requisites, Assumptions – MRP systems and Logic. Materials Budget – Techniques and preparation – Forecasting Techniques and guidelines. Value Engineering: definition, scope and Techniques of value Analysis; value analysis and value Engineering, steps and principles.

MODULE 3: Inventory Control
Functions, classification, determining Inventory levels, Inventory Models and Costs – EOQ – optimum lot size – Economic lot size. ABC analysis and classification – Class A, B and C items – Objective – Limitation of ABC analysis.

MODULE 4: Store Keeping
Functions and Organization – Centralized and de-centralized systems – Role of Store Keeper, essentials of good store keeping. Precaution and security measures in store room. Receipt of Materials – Checking Quality and Quantity; Damage / Storage Report. Use of Bin Cards and Stock register. Inspection of incoming materials – Identification of stores – Material Coding. Store Accounting.

MODULE 5: Purchasing
Organization, Duties, Centralized and decentralized Purchasing – Purchasing Officer – Sources of supply and supplier selection, Buyer – Seller relationship and ethics --- Buying locally and reciprocal buying – Single and multiple source. Legal aspects in buying.

MODULE 6: Purchasing procedure and record
Purchase classification – Requisition, purchase order, follow up and expediting systems. Receipt and inspection.
Records of Purchase – contract, quotation, vendors, records – summary of purchase work and miscellaneous records. Import procedures and documents.

Textbooks/ Reference Books:

1. Materials Management (AICTE: Code No. 333) ----- K. C. Sahu.
2. Materials Management (Forward book depot) ------ C. B. Agarwal.
3. Integrated Materials Management ------ Gopala – Krishna. (Tata McGraw Hills)
4. Purchasing and Materials Management (Tata McGraw Hill) ---- Lee Dobler.

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1818OE22	**Optimization in Operations Research**	**3-0-0**	**3**

Course Outcomes (COs): At the completion of the course the student will be able:
1. To identify technical and managerial situations needing effective, economic and efficient methods for problem solution.
2. To formulate mathematical models for optimization of engineering & business problems for quantitative analysis, solution and interpretation of results.
3. To analyze engineering and managerial problems by classical optimization methods
4. To solve engineering and managerial problems by operations research techniques and interpret the results.

MODULE 1: Introduction
History of Operations Research, Engineering applications, General Statement of an OR problem

MODULE 2: Linear Programming
Introduction to Linear programming, duality, Sensitivity analysis.

MODULE 3: Integer Programming
Introduction to Integer programming; pure integer case, mixed integer case, Cutting plane method, Branch and bound method.

MODULE 4: Analytical methods for Classical Optimization
(a) Formulation of Single and Multi-variate problems with and without constraints
(b) Necessary and sufficient conditions for solving unconstrained problems
(c) Lagrangean method
(d) Karush-Kuhn-Tucker (KKT) conditions and limitations

MODULE 5: Inventory modeling and simulation
(a) Distinction between Deterministic and Stochastic inventory problems, Costs involved in inventory system, Formulation and Solution of simple and basic deterministic inventory problems.
(b) Introduction and meaning of simulation, Monte Carlo simulation and random numbers, Example of simulation and Use of digital computer for simulation.

MODULE 6: Decision Theory
Decision Tree, Game Theory-Minimax and Maximin, Dominance Principle and use of OR software packages

MODULE 7: Queuing Theory
Definitions and Classification, Birth and Death Process, Markovian and semi-Markovian single and multiple server queues, Queuing Networks.

MODULE 8: Dynamic Programming
Principle of Optimality, Concepts of state and stage, Solution of Discrete Problems through Backward Dynamic Programming, Continuous and Multi–stage Dynamic Programming problems.

Textbooks/ Reference Books:

1. Operations Research – H A Taha
2. Operations Research – Gupta and Hira
3. Operations Research – Billy E Gillet
4. Operations Research – Panneerselvam
5. Optimization – S S Rao
6. Operations Research – N G Nair
7. System Simulation by digital computers – N Deo

Course Code	Course Title	Hours per week L-T-P	Credit C
IPE1818OE23	**Management Information Systems**	**3-0-0**	**3**

Course Outcomes (CO): At the completion of the course the student will be able:

1. To *analyze* decision making concept and correlate them with MIS .
2. To *develop* the various MIS and technology of information system in practical life.
3. To *identify and apply* the application of MIS in manufacturing and service sectors.
4. To *design* Database Management System to store various data.

MODULE 1: Management Information System (MIS); Concept, role and impact – organisation – Information characteristics; Taxonomies of information systems. Structure and developments of MIS – Effectiveness.

MODULE 2: MIS basics - Decision making concept and MIS: organization and information, MIS and information: System concept and control: MIS and system concept: System analysis and MIS.

MODULE 3: Development of MIS – Long range plans; class of information and requirement. Implementation of MIS – Procedure, evaluator; Management of Quality in MIS.

MODULE 4: Technology of information system - data, transaction and application processing; TQM for information systems: Human factors and user interface.
Real time systems and design.

MODULE 5: Database Management System – concept, models and design- Conceptual and Physical model. Performance monitoring. MIS AND RDBMS.

MODULE 6: Application of MIS in manufacturing and service sectors. Enterprise management system (EMS) -Enterprise resource planning (ERP). EMS and MIS.

REFERENCE BOOKS:

1. Management Information Systems (Tata McGraw Hill) ------ W.S. Jawadakar.
2. Management Information Systems (BFB Publication) ------ T. Lucey
3. Management Information Systems (Code No. 489, AICTE)----- D.F. Goyal.

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